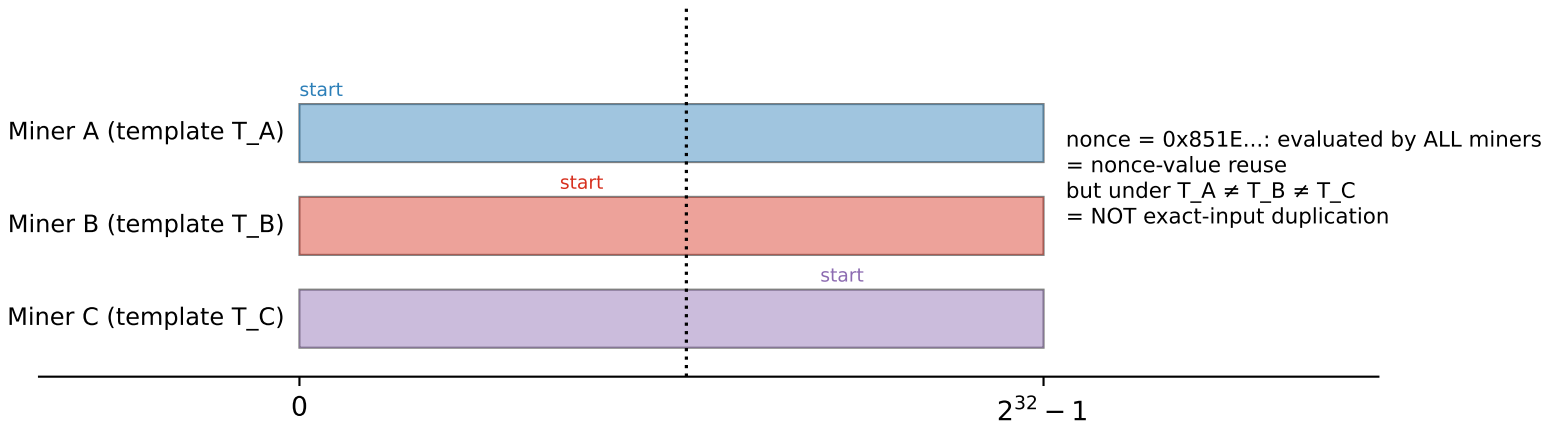


Conventional PoW: independent miners share the same 32-bit nonce-value domain $[0, 2^{32}-1]$



PoCol: deterministic disjoint nonce allocation of the same 32-bit domain

one common immutable template T ; $R_A \cap R_B = R_B \cap R_C = R_A \cap R_C = \emptyset$



within one template epoch: nonce-value reuse = 0 AND exact-input duplication = 0

